

Field operations and source scope

This chapter describes Harbor deployments at disconnected field stations. It supplements the lecture defaults and explicitly changes only the policies it names. The archive schema, observation identity and commit boundary remain the same. A field exception must be read in its stated scope; it is not a global replacement for every classroom rule.

A station may lose catalog connectivity while continuing to collect measurements. Operators need predictable local behavior and an honest record of what was delayed. This chapter follows an observation from collection through reconnection, including the cases where retaining a measurement is safer than routing it with uncertain metadata.

Consider a gateway on campus and a gateway traveling with a field team. Both use observation identifiers and the same durable commit receipt. Only the second is configured for the offline cache exception. If an engineer copies the longer field lifetime into the campus gateway without changing its documented configuration, the engineer has changed the contract. A comparison should therefore begin with a deployment label, then name the operation and its evidence. A useful field notebook records the configuration alongside each trace. That small habit prevents a later reader from treating a valid field result as a contradiction of the classroom lecture. It also allows a test to reproduce the intended policy without guessing from the device's physical location.

Preparing a station

Before departure, the operator copies a signed catalog snapshot and records its epoch. The station checks that every assigned buoy has a mapping. Missing assignments become a preparation failure, not a guessed destination.

The departure checklist separates hardware health from archive permissions. A responsive buoy can still lack authority to submit to a given partition. Keeping these checks separate gives a returning operator a useful explanation for rejected work.

For example, suppose the station is assigned buoys Alder, Birch and Cedar. A departure snapshot contains Alder and Birch but no Cedar row. The preparation process cannot infer Cedar's partition from a nearby identifier or from the previous trip. It marks the assignment incomplete and asks for a corrected snapshot. A second check verifies the snapshot signature and epoch before the station accepts even the two present rows. Passing the mapping check still does not establish that a lease will be valid later. Permission can expire between preparation and collection. Treat the checklist as evidence about readiness at a particular time, and preserve later lease checks at the moment a write is attempted.

Collection without connectivity

A disconnected buoy appends measurements to a local spool before reporting them as retained. Each spool entry keeps its original observation identifier and event timestamp. Reconnection changes the transport attempt, not the identity of the physical observation.

A spool checksum detects damaged local storage. It does not prove that an observation reached the archive. Operators compare durable commit receipts after reconnecting, and delete local entries only under the documented acknowledgment rule.

Imagine observation O-314 is sampled while the radio is offline. The buoy records its bytes and checksum locally before displaying retained. When the link returns, attempt A-1 sends O-314 but times out. Attempt A-2 must still refer to O-314. The local interface may show two transport attempts and one retained measurement; that is consistent. If A-2 receives a durable receipt, the buoy can mark the spool entry confirmed. If the spool checksum fails before transmission, the operator instead records local corruption. A checksum failure is not an archive rejection, because the archive may never have received the observation. Keeping these states distinct is essential to a report that accurately describes where data was lost or delayed.

The field spool boundary

The field spool holds up to 240 observations per buoy. When full, the buoy stops accepting new scheduled readings and records a capacity alarm. It never silently overwrites the oldest unacknowledged observation.

A larger spool tolerates a longer outage but cannot make an indefinite outage harmless. Capacity planning uses the measurement rate and an outage assumption. If either changes, the predicted time to exhaustion changes too.

At two scheduled readings per minute, an empty 240-entry spool provides 120 minutes of capacity under the simplified assumption that no entries are confirmed during the outage. If it already contains 60 unconfirmed entries, only 180 slots remain, providing 90 minutes at the same rate. This arithmetic estimates admission capacity; it does not guarantee battery life, radio recovery or storage health. An operator should recompute it when the schedule changes. A stop at the capacity limit is visible evidence of bounded behavior. Overwriting old entries would make the system appear to keep collecting while quietly destroying measurements whose archive outcome is still unknown. The capacity alarm makes that tradeoff explicit to the person responsible for recovery.

Reconnect in bounded batches

A station reconnects in batches of twenty observations. It waits for the batch outcome before admitting another batch from the same buoy. Other buoys can progress independently when gateway capacity permits.

The batch size limits the uncertainty surface of one interrupted exchange. It does not create a transaction across all twenty records. Each observation still has its own commit receipt and replay identity.

Suppose a batch contains O-1 through O-20. The archive commits the first seven records before the connection fails. A returning connection must resolve each identifier, rather than interpreting batch timeout as twenty failed commits. The seven existing records can return original receipts. The remaining identifiers can proceed under the normal transaction rules. The next batch for that buoy waits until these outcomes are known. A second buoy can still make progress if its own batch is ready and gateway capacity is available. This structure limits one buoy's outstanding uncertainty without imposing a global pause on every sensor. Test partial completion deliberately; an all-successful batch does not demonstrate correct behavior when the response is interrupted halfway through.

Preserve event time

Replayed field measurements preserve the event time assigned when the sensor sampled them. The gateway adds a separate arrival timestamp. Overwriting event time with reconnection time would move old observations into the wrong stream window.

Late handling is a policy choice after timestamp preservation. A late ledger can retain the evidence even when the published aggregate is final. This avoids turning delayed delivery into a false claim about when the world was measured.

A sample taken at event-time second 42 may arrive after reconnection at second 95. Its event-time window remains the one starting at 40. Assigning it to the window starting at 80 would describe arrival traffic instead of environmental measurements. Both reports might be useful, but they answer different questions and need different names. The raw record therefore retains both timestamps. A window processor can use event time for the measurement aggregate and arrival time for an operational ingestion chart. During review, compare a few individual identifiers across the spool, archive and aggregate. This makes a timestamp substitution visible even when totals happen to be similar. Equal daily counts alone cannot prove that records were assigned to the correct windows.

Receipts and uncertain outcomes

A lost response during reconnection leaves the buoy uncertain about remote commit. It asks about the original observation identifier and accepts the original receipt if one exists. Creating a fresh observation identifier would remove the archive's ability to recognize a repeat.

The six-hour replay table from the transaction lecture limits automatic recognition. A long field outage therefore requires a separate reconciliation procedure instead of assuming that every old identifier remains in the online replay table.

Walk through a timeout after the archive has acknowledged both journal copies but before the response reaches the buoy. The durable record exists; the buoy's uncertainty is local. Within the replay interval, querying with the original observation identifier recovers the original receipt. If a developer instead constructs a new identifier for the retry, the archive sees a different proposed observation and cannot apply the intended duplicate rule. A regression should make the first response disappear while retaining the committed record. It then inspects the number of durable observations after retry. Merely checking that the second request returned success would let the defect pass, because both a correct replay and an incorrect duplicate append can return a successful transport response.

Catalog snapshots

A field snapshot is a complete mapping for the assigned buoy set at one epoch. The station does not merge arbitrary rows from different epochs. A partial refresh stays pending until the station can validate the replacement snapshot.

This rule favors explainable consistency over partial progress. The operator can identify exactly which catalog version routed a measurement. Combining half of one revision with half of another would make that explanation ambiguous.

Assume epoch 18 maps Alder to partition West and Birch to partition East. Epoch 19 swaps those assignments. A station that combines Alder from epoch 18 with Birch from epoch 19 would route both to West, even though neither complete snapshot says to do so. The field snapshot rule rejects that mixed state. Downloading a candidate can occur in pieces, but activating it is a separate operation after completeness and integrity checks. Retain the old active snapshot while a replacement is incomplete, subject to its ordinary freshness boundary. If the old snapshot expires before activation succeeds, spool the new work. The system should expose that temporary lack of routing authority rather than conceal it behind a partially assembled catalog.

Offline cache exception

For offline field stations, the mapping cache lifetime is one hundred twenty seconds after loading. This replaces the forty-five-second standard lecture lifetime only in the field-station configuration. At the boundary, a new route requires a successful refresh.

The longer lifetime is a deployment tradeoff explicitly chosen for intermittent links. It is not permission to reuse a mapping forever when the catalog is unreachable. The station spools observations if it cannot obtain a fresh authorized route.

A field entry loaded at wall-clock second 100 remains within its lifetime at second 219 and reaches its boundary at second 220. A new route at 220 requires refresh. A standard lecture entry loaded at the same time reaches its boundary at second 145. These examples use wall-clock seconds, not logical log ticks. The comparison explains why a question must identify the deployment. The exception grants a longer valid interval after a successful load; it does not reset the load time on every cache hit. A sliding implementation that extended expiry with every read would allow a popular stale mapping to persist indefinitely. Test repeated hits near the boundary to ensure they do not silently renew its authority.

Freshness under revision

Receiving a higher catalog epoch invalidates the older mapping immediately, even if its time-based lifetime has not elapsed. The longer field lifetime changes the timeout fallback; it does not disable revision ordering.

A test must include both ways a mapping becomes unusable. One trace advances time to the lifetime boundary. Another supplies a newer epoch while the cached entry is still young. Both should route through refresh or explicit failure.

Consider a field cache entry loaded at second 100 with epoch 18. At second 130, the station receives a verified epoch-19 snapshot. Although the old entry would otherwise remain young, the newer revision takes precedence immediately. Keeping epoch 18 until its timeout would discard the meaning of the invalidation. Conversely, a late response carrying epoch 17 cannot replace epoch 19 merely because it arrived later in wall time. The implementation compares revision before accepting the response. This is a useful example of two clocks serving different purposes: elapsed time bounds how long information may be reused, while epoch orders authoritative revisions. A trace should contain both fields so an engineer can tell which rule caused the refresh or rejection.

Reconciliation after six hours

After more than six hours without a confirmed receipt, the field operator uses an archive reconciliation report keyed by buoy and observation identifier. The operator does not assume the short-lived replay table still contains every record.

Reconciliation is an explicit operational path with an audit record. It can identify previously committed observations even after the fast replay cache expires. The operator must resolve payload conflicts before resubmitting ambiguous records.

A buoy returns after an eight-hour outage with an unconfirmed entry. The online replay table may have discarded its fast lookup, but the durable archive can still contain the committed observation. A missing replay-cache entry is therefore insufficient evidence for a fresh append. The reconciliation report searches the durable identity record and returns an existing receipt, a definitive absence or a conflict needing attention. Those outcomes drive different next steps. Preserve the report identity in the operator audit so a later reviewer can reconstruct why resubmission was allowed. The slower path is deliberately explicit because it handles uncertainty beyond the ordinary online shortcut. Its existence does not extend the advertised six-hour fast replay interval or remove the need to preserve identifiers.

Separating alarms

A capacity alarm means the local spool cannot admit more readings. A routing alarm means a fresh mapping is unavailable. An archive alarm means an attempted storage operation failed or remained uncertain. These alarms can occur together but require different actions.

Grouping all three as upload failure obscures where evidence lives. The spool may still safely hold measurements during a routing alarm. Conversely, a healthy route is little comfort if the local spool has already stopped accepting new readings.

Suppose the catalog is unreachable while 100 measurements remain safely in the spool. The appropriate first diagnosis is routing unavailable, not lost measurements. If the outage continues until the spool fills, a capacity alarm adds a new fact: future scheduled readings can no longer be admitted. A later archive busy response creates yet another category once routing resumes. A status panel should allow these facts to coexist instead of replacing every prior alarm with the newest one. The recovery order follows dependencies: obtain a valid route, resolve archive outcomes and drain retained work, then restore normal collection admission when capacity permits. Keeping a timeline of these transitions helps distinguish data retained during an outage from readings never accepted after the spool became full.

Interpreting a delayed report

An analyst reading a report checks window identity, revision and finalization status. A field measurement that arrives after the correction boundary belongs in the late ledger. Its absence from the finalized aggregate is then a documented policy outcome.

This distinction prevents an analyst from interpreting every difference between raw archive data and a report as arithmetic corruption. The report represents a defined window and correction policy, while the raw archive preserves individual observations.

Take a twenty-second window starting at 40. Its normal end is 60, and the lecture correction rule extends acceptance through its stated watermark boundary. A measurement received after finalization remains a valid individual archive record but does not necessarily belong in the already published aggregate. An analyst can produce a separate late-data report if the task calls for it. The report must identify that additional scope rather than quietly rewriting a finalized result. When comparing two dashboards, check whether one displays current revisions and the other pins a finalized revision. Apparent disagreement may then be a difference in report contract. It should still be investigated with identifiers and timestamps, not dismissed merely because late delivery is possible.

A supervised lantern repair

A lantern lease is available only during a supervised repair session. The supervisor records the partition and holder before the gateway writes. The special renewal limit is documented in the lease lecture's speaker notes and is not restated here.

The normal fencing generation remains mandatory. A supervised process can still pause and resume with stale authority. Human oversight adds accountability but does not remove the need for an archive-side generation check.

During a repair, a supervisor may see a gateway respond normally and assume its authority is still current. That observation proves only that the process is alive. The archive must inspect the lease generation on each durable write. If ownership transferred while the gateway was paused, its old generation is rejected even when the repair operator has not noticed the transfer. The special renewal policy also remains bounded; supervision does not grant unlimited extensions. To answer a question about the exact lantern renewal, consult the named lecture notes rather than infer a value from the standard lease duration. This intentional split illustrates a realistic reading task in which a chapter refers to a policy whose complete definition lives in a lecture annotation.

Audit a reconnect

The reconnect audit names the station, buoy, batch, observation and outcome. It avoids reusable tokens and raw measurement payloads. Engineers use these fields to correlate local spool entries with archive receipts without exposing credentials.

An audit that records only batch success loses information when a batch partially completes. Per-observation outcomes support selective replay and make it possible to explain why one local entry was deleted while another was retained.

For a partially successful batch, an audit might record that O-1 received its original receipt, O-2 was newly committed and O-3 remained uncertain after a timeout. The payload values are unnecessary for explaining those outcomes. A report-level request identifier can link the events without exposing the credential used for the network exchange. During review, verify that logging an exception does not accidentally serialize the entire request object, including headers. A minimized success log is insufficient if the failure path dumps secrets. The same boundary applies to temporary diagnostic views: useful source coordinates and timings can be displayed without retaining a user's access token. Audit quality is measured by explanatory value and deliberate scope, not by the quantity of copied data.

Measure useful throughput

Field throughput counts newly confirmed observations per second. Replaying an already committed identifier can produce a successful response without adding a new record. Counting every successful HTTP response would overstate newly stored work.

Track replay receipts separately from new commits. Both represent useful progress toward emptying the local spool, but they answer different questions about archive load and physical measurements.

Suppose a reconnect receives 100 successful responses in ten seconds. Forty responses are original receipts for records already committed, and sixty represent newly committed observations. The response rate is ten per second, while the new-commit throughput is six per second. The spool can still make useful progress on all 100 confirmed entries. Neither metric is false, but they must be named accurately. If a performance change increases repeated retries, response rate may rise while useful new storage stays constant or falls. A benchmark should therefore report attempt count, original replay receipts and new durable commits together. This prevents an optimization claim from being driven by duplicated transport activity instead of the workload the archive was intended to perform.

Restoring station metadata

After a station crash, restore the spool index from its verified local checkpoint and inspect entries after the checkpoint boundary. Never infer archive absence solely from a missing local receipt. Reconcile uncertain identifiers before starting a new append.

The restore procedure has two evidence sources: local retained measurements and archive durable outcomes. Neither one alone necessarily describes the complete history of an interrupted network exchange.

Imagine the station checkpoint records O-51 as unconfirmed. After that checkpoint, the archive commits O-51 and returns a receipt, but the station crashes before persisting its local confirmation. Restoring the checkpoint brings back an apparently pending entry. Resubmission with the same observation identifier can recover the receipt; assigning a fresh identifier would confuse recovery with a new measurement. Now consider the opposite case, where the station wrote a local intent but never reached the archive. The same restore path must allow definitive absence to trigger a valid append attempt. Test both histories around the crash boundary. A restore procedure that handles only already-confirmed rows may look correct during a clean shutdown and fail precisely when its recovery function is needed.

Design a reproducible drill

Choose a small synthetic buoy set and fixed observation identifiers. Record the intervention boundary, such as disconnecting immediately after archive commit. Preserve expected observable outcomes separately from the test materials given to the system.

A drill should be capable of failing for a realistic defect. For example, a client that assigns a new identifier on every timeout should create a duplicate under a lost-response scenario. A check that only asserts the process remains alive would miss that defect.

A useful drill has an arrangement, an intervention and an observable claim. Arrange one buoy with a retained observation and an empty archive. Intervene by dropping the response after the durable commit. Then assert one archive observation and an original receipt after replay with the same identifier. Preserve a deliberately broken client that changes the identifier, or otherwise show that the assertion detects the resulting second record. This calibration distinguishes a meaningful check from a script that prints success regardless of behavior. The test's expected result belongs in its independent oracle, outside the course materials or model prompt. Otherwise an assistant could repeat the expected wording without demonstrating that ingestion, retrieval or recovery actually supplied the evidence needed to derive it.

Compare source scopes

When a lecture and field chapter give different constants, first identify the deployment named in the question. The standard cache lifetime and the offline field lifetime are deliberate scoped policies. Reporting one universal number would erase the distinction.

A good technical answer can state both policies and explain when each applies. It should cite the location that actually contains the selected number, rather than a neighboring passage that merely discusses caches.

An answer saying the cache lasts 120 seconds is incomplete when the question compares the lecture and field chapter. The reader needs two numbers with two scopes and the relevant coordinates. An answer saying the sources disagree without explaining the deployment exception is also incomplete: the difference is deliberate and reconcilable. This is a useful pattern for technical reading more broadly. Specifications often contain a default, an exception and a boundary on that exception. Summaries should preserve all three when they affect the decision. Source selection matters as much as fluent prose. A nearby paragraph about cache performance cannot support a precise lifetime merely because it contains the word cache; the cited excerpt must contain or directly establish the claimed rule.

Stop conditions

An operator stops automatic replay on a payload conflict, a failed catalog integrity check or a revoked write lease. Transient archive busy responses use the bounded retry schedule from the transaction lecture.

Stopping is part of correct recovery. Repeating a request indefinitely can hide a permanent contradiction and consume resources needed for valid work. The incident record should preserve the first decisive failure and the action needed to resolve it.

Compare three responses to a reconnect attempt. Archive busy is transient and can use the bounded waits from the transaction lecture. Payload conflict means the same observation identifier is associated with different content and needs explicit resolution. Revoked lease means the gateway lacks current authority and must not continue writing under the old lease. Treating all three as a generic retryable error would blur a resource delay, a data contradiction and an authorization boundary. A recovery client should preserve the first decisive classification and expose a concrete next step. It should also count total attempts carefully: a policy allowing three retries after the initial attempt permits four attempts, not three and not an unbounded loop until something eventually succeeds.

The recovery shelf

Field stations retain three verified metadata checkpoints, matching the archive checkpoint count. The shelf removes the oldest verified checkpoint only after the new candidate passes verification. An incomplete candidate cannot evict a usable recovery point.

Verification should read the candidate through the same restoration path used after a crash. Merely observing that a file was written does not prove that its contents can recreate the intended state.

Suppose verified checkpoints C-7, C-8 and C-9 are on the shelf when candidate C-10 is written. If C-10 fails checksum validation, the usable set remains C-7 through C-9. Deleting C-7 immediately when the candidate file appeared would reduce the recovery set before its replacement became trustworthy. When C-10 passes the real restoration check, the shelf can retire C-7 and retain C-8 through C-10. The policy counts verified recovery points, not filenames. A drill should fail candidate validation deliberately and inspect the remaining shelf. This is another case where ordering matters: writing, verifying and deleting are separate transitions, and a successful final count alone may hide a period in which the last usable recovery point was exposed to unnecessary risk.

Regional maintenance window

The fictional North Quay station schedules routine compaction when no supervised repair lease is active. Compaction preserves observation identifiers and committed outcomes. It can reclaim obsolete index pages without rewriting the meaning of a receipt.

Operators keep maintenance events separate from client retries in the trace. A slowdown during compaction can produce timeouts, but the ordinary uncertain-outcome rules still apply. Maintenance does not authorize fresh observation identifiers for old measurements.

Compaction can relocate records and rebuild index pages while preserving their logical identities. An engineer reviewing a post-compaction receipt should still find the same observation and committed outcome. Physical file offsets are unsuitable as permanent observation identifiers because maintenance may change them. During a drill, sample receipts from before the maintenance window and resolve them afterward. Also measure timing separately from correctness: a longer lookup during compaction can be an expected performance effect, while a missing committed observation is a different class of defect. The absence of a supervised repair lease is a prerequisite for the routine North Quay operation, not proof that every other dependency is healthy. Keep the maintenance checklist explicit rather than letting a single green condition stand for complete readiness.

The glacier override

At North Quay only, recovery from a checksum mismatch requires the operator to select checkpoint label glacier-27 and wait seventeen seconds before replaying the journal. This station-specific drill rule appears in this chapter and has no standard lecture equivalent.

The pause gives the laboratory harness time to verify its isolated storage mount. It is a fictional course rule, not general disaster-recovery advice. If glacier-27 is unavailable, the operator stops the drill and reports a missing prerequisite instead of inventing a replacement.

For the North Quay exercise, an operator first identifies a checksum mismatch and confirms that the station-specific recovery procedure applies. The operator then selects the named checkpoint and starts a fresh measured wait before journal replay. A generic latest-checkpoint routine is insufficient for this drill because it could choose a different label. Similarly, recording that a pause occurred is weaker evidence than recording the start and replay times. If the required checkpoint is missing, stopping preserves the meaning of the exercise. Substituting a convenient candidate would test another procedure. The rule is intentionally located late in the chapter so a retrieval system must ingest and search the complete reading instead of succeeding from only an opening summary or the first few pages.

A final evidence record

A completed field drill records original file identities, observed extraction or import outcomes, the chosen source coordinates, and the behavior after recovery. It distinguishes predicted results from measured ones and keeps the remaining limitations visible.

This chapter has described a coherent local policy rather than a universal distributed-systems specification. The useful habit is to carry scope, identity and evidence through an explanation. That habit remains valuable when the particular course constants change.

A final report might state that all course files imported, the local semantic index returned the intended lecture note, and the remote answer request timed out. Those observations support a successful local ingestion and retrieval claim but leave the answer claim unqualified. Another run might return a fluent answer with the wrong source coordinate; that is a separate defect even when the answer's number happens to be correct. The reader should be able to tell which outcome occurred without reconstructing hidden tool logs. Include the artifact identity, exact question, actual cited coordinate and remaining uncertainty. Preserve course originals across cleanup while removing temporary imported copies and private traces according to the session contract. Completion means the stated observable objective was met, rather than that a script merely reached its last line.